SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: **MLA03**
COURSE NAME: Reinforcement learning for Environment and Agent

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COURSE OUTCOME COVERED

CO4: *Evaluate model-based reinforcement learning approaches for prediction, planning, and policy optimization. (BL3, BL4, BL5)*

Assessment Tool Weightage: 30%

Dr G. A. SENTHIL

QUESTIONS: 10 **Total Marks: 50**

Annexure A

Technical Presentation **DURATION: 2HRS**

1. **Dynamic Programming for Sequential Decision-Making in Reinforcement Learning**
    - Bellman Principle of Optimality
    - Policy Evaluation and Policy Improvement
    - Value Iteration vs. Policy Iteration
    - Industrial applications

(25 Marks)

2. **Monte Carlo Methods and Temporal-Difference Learning: A Comparative Study**
    - Monte Carlo Prediction and Control
    - TD(0), SARSA, and Q-Learning
    - Sample efficiency
    - Performance comparison through experiments

(25 Marks)

3. **Deep Q-Networks for Intelligent Decision-Making**
    - DQN Architecture
    - Experience Replay
    - Target Networks
    - Industrial applications in robotics and autonomous systems

(25 Marks)

4. **Comparative Analysis of Advanced Deep Q-Learning Algorithms**

- Double DQN (DDQN)
- Dueling DQN
- Prioritized Experience Replay (PER)
- Performance comparison using benchmark environments

(25 Marks)

5. **Policy-Based Reinforcement Learning for Continuous Control Problems**
   - Vanilla Policy Gradient
   - REINFORCE Algorithm
   - Stochastic Policy Search
   - Applications in robotics and autonomous driving

(25 Marks)

6. **Actor-Critic Methods for Real-Time Intelligent Systems**
   - Advantage Actor-Critic (A2C)
   - Asynchronous Advantage Actor-Critic (A3C)
   - Performance evaluation
   - Industrial automation applications

(25 Marks)

7. **Advanced Policy Optimization Techniques in Deep Reinforcement Learning**
   - Proximal Policy Optimization (PPO)
   - Trust Region Policy Optimization (TRPO)
   - Deep Deterministic Policy Gradient (DDPG)
   - Comparative analysis of convergence, stability, and efficiency

(25 Marks)

8. **Model-Based Reinforcement Learning for Autonomous Decision-Making**
   - Model-Based RL Framework
   - Environment Modelling
   - Planning Strategies
   - Applications in autonomous robots and smart manufacturing

(25 Marks)

9. **Sampling-Based Planning and Model-Based Policy Optimization**
   - Sampling-Based Planning
   - Model-Based Data Generation
   - Value-Equivalence Prediction
   - Model-Based Policy Optimization
   - Real-world industrial case studies

(25 Marks)

10. **Industrial Applications of Advanced Reinforcement Learning**
    - Smart Manufacturing
    - Autonomous Vehicles
    - Healthcare Decision Support
    - Smart Grid Energy Management
    - Wireless Communication and Network Optimization
    - Finance and Algorithmic Trading

- Comparative analysis of Value-Based, Policy-Based, and Model-Based RL approaches

(25 Marks)

Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness